

Olasentra Event Operations Command Platform

Database Health Report

Live production audit — read-only analysis

6 June 2026 · olastofx_eventstaff · MariaDB 11.4.10

Post Phase 16 · Gate G monitoring period



Executive Summary (Business)

Overall verdict: Database is **healthy for current scale** (~63 staff, 392 registrations, 32 events). Phase 16 verified: zero orphans, FKs in place, indexes applied.

Score	Rating	Summary
82	Overall Health	Solid live ops; good backup discipline
78	Performance	Fine today; tune before 2,000+ staff
68	Security	FKs improved; secrets in DB settings = main risk
90	Storage	Tiny DB (~0.3 MB); site files dominate disk
85	Data Quality	Clean emails; multi-event regs intentional

Priority issues

Severity	Issue	Impact
Critical	API keys, SMTP password, OAuth tokens in system_settings	Credential leak if backup exposed
High	deleteStaffProfileCompletely() incomplete cascade	GDPR erasure gap
High	getAllEvents() aggregates all registrations per load	Dashboard slows at scale
Medium	LOWER(TRIM(email)) bypasses indexes	Search slows at 5k+ rows
Medium	system_settings MyISAM + latin1	Engine/charset inconsistency
Low	5 platform ops tables (P18–P22) + 3 dormant equipment tables	Equipment rental out of scope

Section 1 — Database Overview

Metric	Value
Database name	olastofx_eventstaff
Server	MariaDB 11.4.10-cll-lve (shared hosting)
Total tables	27 (19 with data + 8 empty platform P51)
Total records	~1,045 rows
Database size (dump)	326.5 KB (~303 KB data payload)
Index size (est.)	80–120 KB (~30–40% of data)
Storage engines	InnoDB: 17 tables · MyISAM: 2 (system_settings, push_subscriptions)
Live health	app ok · database ok · storage ok

Largest tables

Rank	Table	Rows	Data KB	% data
1	staff_registrations	392	162.1	53.5%
2	app_notifications	210	63.2	20.9%
3	staff	63	26.2	8.7%
4	admin_audit_log	154	16.2	5.4%
5	system_settings	69 keys	15.1	5.0%

Fastest growing

Table	Signal	Rate est.
staff_registrations	392 rows in ~4 days	~200–400/week peak season
app_notifications	210 JSON payloads	~50/day active ops
admin_audit_log	154 entries	~35/day
ip_geo_cache	107 lookups	Grows with traffic

Section 2 — Table Analysis (all 27 tables)

Table	Purpose	Rows	KB	Health	Notes
staff_registrations	Per-event registrations	392	162	A	53% of DB
app_notifications	Notification feed	210	63	A-	Fast growth
staff	Canonical profiles	63	26	A	Unique email
admin_audit_log	Admin audit	154	16	A	Good coverage
events	Event master	32	8	A	Summer roster
system_settings	Config + secrets	69	15	C	MyISAM; secrets
ip_geo_cache	Geo cache	107	8	B+	TTL cleanup
venues	Venues	7	1	A	Reference
work_types	Work types	5	1	A	Seed
personal_invoice_lines	Invoice lines	3	<1	A	FK P16
saved_job_records	Job records	2	<1	A	FK P16
admin_users	Admin accounts	1	<1	B	Single user
attendance	Check-ins	0	0	A	Pre-event
commission_invoices	Commission	0	0	A	Unused
commission_invoice_lines	Comm. lines	0	0	A	Unused
staff_messages	Inbox	0	0	A	FK P16
staff_blacklist	Blocked emails	0	0	A	Empty OK
push_subscriptions	PWA push	0	0	B	MyISAM
website_visits	Analytics	0	0	A	Enabled
platform_cleanup_log	P18	0	0	A	Schema only
platform_storage_snapshots	P19	0	0	A	Schema only
platform_audit_runs	P20	0	0	A	Schema only
platform_ssl_checks	P21	0	0	A	Schema only
emergency_event_log	P22	0	0	A	Schema only
equipment_categories	—	0	0	N/A	Out of scope — no dev
equipment_items	—	0	0	N/A	Out of scope — no dev
equipment_rentals	—	0	0	N/A	Out of scope — no dev

Empty operational: attendance, commission_*, staff_messages, staff_blacklist, push_subscriptions, website_visits · **Archive candidates:** ip_geo_cache (>90d), admin_audit_log (> 12mo)

Section 3 — Index Analysis

Post-Phase 16 indexes (applied)

- `idx_staff_reg_event_status` on staff_registrations (event_id, status)
- `idx_events_active_date` on events (is_active, event_date)
- `idx_staff_email_lower` on staff (email)
- `idx_notif_admin_feed` on app_notifications (audience, is_read, created_at)

Finding	Severity	Detail
Duplicate indexes	Medium	staff.email: UNIQUE + idx_staff_email + idx_staff_email_lower
Overlapping indexes	Low	idx_staff_event + idx_staff_status vs composite
Missing functional index	Medium	LOWER(TRIM(email)) queries bypass btree
Full scan risk	Medium	getAllEvents() GROUP BY all registrations

Table	Query	Suggested index	Improvement
staff_registrations	LOWER(TRIM(email)) lookup	email_normalized generated column + index	10× at 5k+ rows
staff	Same pattern	Drop redundant idx_staff_email	Write efficiency
events	Active upcoming	idx_events_active_date	Indexed
app_notifications	Inbox feed	idx_notif_admin_feed	Indexed

Section 4 — Query Performance

Area	Query pattern	Cost now	Cost @ 5k staff	Recommendation
Dashboard	COUNT pending registrations	Low	Low	Composite index
Staff list	event_id + status filter	Low	Low	P16 index
Staff directory	LOWER(TRIM(email))	Medium	High	Normalized email column

Events	getAllEvents() GROUP BY	5–15ms	50–150ms	Cache registration counts
Attendance	Token UNIQUE lookup	O(1)	Low	✔ checkin_token
Sheets sync	Per-event API loops	Medium	High	Batch sync

Section 5 — Orphan Record Analysis

All checks **zero** post-Phase 16 deploy (verified 2026-06-06 14:44 UTC).

Check	Count	Severity	Action
missing staff_id	0	—	✓ Phase 49 backfill
orphan staff_id	0	—	✓
orphan staff_messages	0	—	✓ FK P16
orphan invoice_lines	0	—	✓ FK P16
orphan job event_id	0	—	✓ FK P16
attendance orphans	0	—	Empty table

Section 6 — Data Quality Report

Check	Result
Duplicate staff emails	0 (63 unique / 63 staff)
Duplicate per-event email	0 (uq_staff_email_event)
Multi-event registrations	63 people → 392 regs (avg 6.2 events/person) — by design
Staff ↔ registration link	✓ Phase 49 complete
PSA fields	Nullable on staff — profile gate enforces
Invalid event refs	0 orphan event_id
system_settings charset	latin1 MyISAM — migrate recommended
Secrets in settings	Present — Phase 15 target

Data Quality Score: 85 / 100

Section 7 — Storage Analysis

Item	Size
Live DB dump	326.5 KB
On-disk est. (data + indexes)	400–450 KB
site-files.zip (weekly)	431.8 MB (1,552 files)
settings-and-cms.json	24.1 KB

Horizon	DB size est.	Driver
3 months	1.5–2.5 MB	Registrations + notifications
6 months	4–6 MB	+ audit + attendance
12 months	10–15 MB	Notifications JSON + audit

Section 8 — Cleanup Opportunities

Item	Category	Savings
ip_geo_cache > 90 days	Safe to remove	5–50 KB
admin_audit_log > 12 months	Review required	Grows over time
Rejected regs > 24 months	Review required	Policy dependent
Application log files	Safe to remove	MB–GB (filesystem)
Empty platform tables	Do not remove	P18–P22 foundation
equipment_* tables	Review later	May drop if approved — not in product roadmap
Duplicate staff indexes	Review required	10–20 KB

Section 9 — Security & Compliance

FK relationship	Status
staff_registrations → staff	✓
staff_messages → staff	✓ P16
personal_invoice_lines → saved_job_records	✓ P16
saved_job_records → events, admin_users	✓ P16
attendance → staff_registrations	⚠ No FK
app_notifications	⚠ Polymorphic, no FK

Section 10 — Operations Readiness

Module	Score	Blockers
Command Center	78%	getAllEvents() aggregation
Event Hub	82%	Ready
Live Ops	55%	attendance empty; emergency schema ready
Unified Inbox	60%	0 messages; delete cascade incomplete
Auto Approval	50%	No rules engine in DB
Staff Intelligence	40%	Trust scores table not created
Client Portal	75%	63 profiles; PSA fields present
AI Operations Assistant	15%	No embeddings/log tables

Section 11 — Future Scale Report

Scale	Regs est.	DB est.	Bottleneck
500 staff	3,000–4,000	2–3 MB	getAllEvents() GROUP BY
1,000 staff	6,000–8,000	5–8 MB	LOWER(TRIM email)
5,000 staff	30,000–40,000	30–50 MB	Dashboard + Sheets sync
10,000 staff	60,000–80,000	60–100 MB	Shared hosting query time

Section 12 — Action Plan

Priority	Action	Phase
Critical	Externalize secrets from system_settings	P15
High	deleteStaffProfileCompletely cascade fix	P16b
High	Cache event registration counts	P2 Command Center
Medium	Normalized email index	P16b+
Medium	Audit log retention policy	P18
Low	Drop duplicate idx_staff_email	After EXPLAIN

30-day plan

1. Gate G → Phase 15 security (secrets)
2. Phase 16b: runner + delete cascade
3. Event count cache + EXPLAIN audit
4. P18 Cleanup: geo TTL + audit rotation

Appendix A — SQL Recommendations (do not execute without approval)

```
-- DIAGNOSTIC (read-only)
SELECT table_name, engine, table_rows,
       ROUND(data_length/1024,1) AS data_kb,
       ROUND(index_length/1024,1) AS index_kb
FROM information_schema.tables
WHERE table_schema = DATABASE()
ORDER BY data_length DESC;

EXPLAIN SELECT COUNT(*) FROM staff_registrations
WHERE event_id=1 AND status='pending';

-- ORPHAN RE-CHECK (expect 0)
SELECT COUNT(*) FROM staff_registrations
WHERE staff_id IS NULL OR staff_id=0;

-- OPTIMIZATION (change window)
ALTER TABLE staff_registrations
  ADD COLUMN email_normalized VARCHAR(150)
  GENERATED ALWAYS AS (LOWER(TRIM(email))) STORED,
  ADD INDEX idx_sr_email_normalized (email_normalized);

-- CLEANUP (P18 — review first)
DELETE FROM ip_geo_cache
WHERE cached_at < DATE_SUB(NOW(), INTERVAL 90 DAY);
```

Appendix B — Methodology

- Production mysqldump 2026-06-06 14:43 UTC (326.5 KB) for row/size stats
- Phase 48–51 verified via deploy log 14:44 UTC
- Codebase query review (staff-repository, events-repository, attendance)
- No live information_schema or production SQL executed
- Index sizes estimated; no slow query log available

